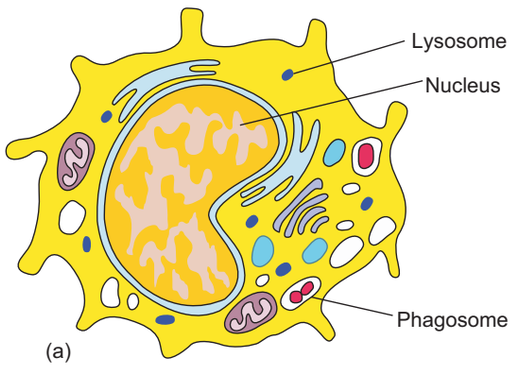
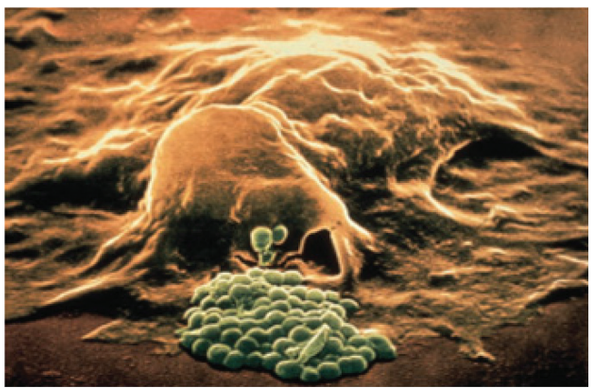
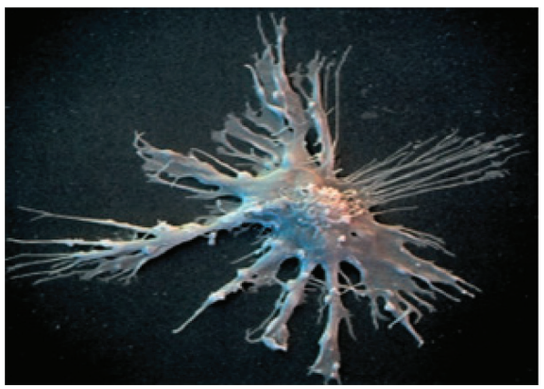




(b)



(c)

Figure 13.12: (a) Structure of Monocytes (b) Phagocytosis by a Macrophage (c) Dendritic Cell

hours, enlarge, migrate to the tissues and mature into macrophages or dendritic cells (Figure 13.12 a).

Macrophages are derived from monocytes and are classified as mononuclear phagocytic leukocytes. These microbial molecules are examples of pathogen associated molecular patterns (PAMPs) (Figure 13.12 c).

PAMPs enable macrophages to distinguish between potentially harmful microbes and other host molecules. After the pathogen is recognized, the macrophages', pattern recognition receptors (Example: Toll like receptors) bind the pathogen and phagocytose it. Macrophages also have receptors for antibodies and complement proteins. Both antibody and complement proteins can coat microorganisms and enhance their phagocytosis. This enhancement is termed opsonization. Macrophages spread throughout the body and take up residence in specific tissues. Macrophages serve different functions in different tissues and are named according to their tissue location.

- Alveolar macrophages in the lung
- Histiocytes in connective tissue
- Kupffer cells in the liver
- Mesangial cells in the kidney
- Microglial cells in the brain
- Osteoclasts in bone

d. Dendritic cells

Dendritic cells are not a single cell type. They are a heterogeneous group of cells so named because of their Dendron (neuron) like appendages (Figure 13.12d). They arise

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